



Real IEA data analysis: The explosive demand for critical minerals driving the clean energy transition.
From copper to lithium, explore the unprecedented supply chain challenges ahead.

Analysis Controls

Scenario

Stated Policies

Announced Pledges

Net Zero 2050

Select Mineral

Copper

View Type

 Demand Trends

 2024 vs 2050

 EV Minerals

 Total Demand
2024

13M

Kilotons across all
minerals

 Total Demand
2050

43M

Kilotons in Net Zero
2050

 Total Growth

3.4x

Demand multiplier by
2050

 Copper
Dominance

2.5x

Copper growth (8M →
20M kt)

 Lithium EV
Surge

14.2x

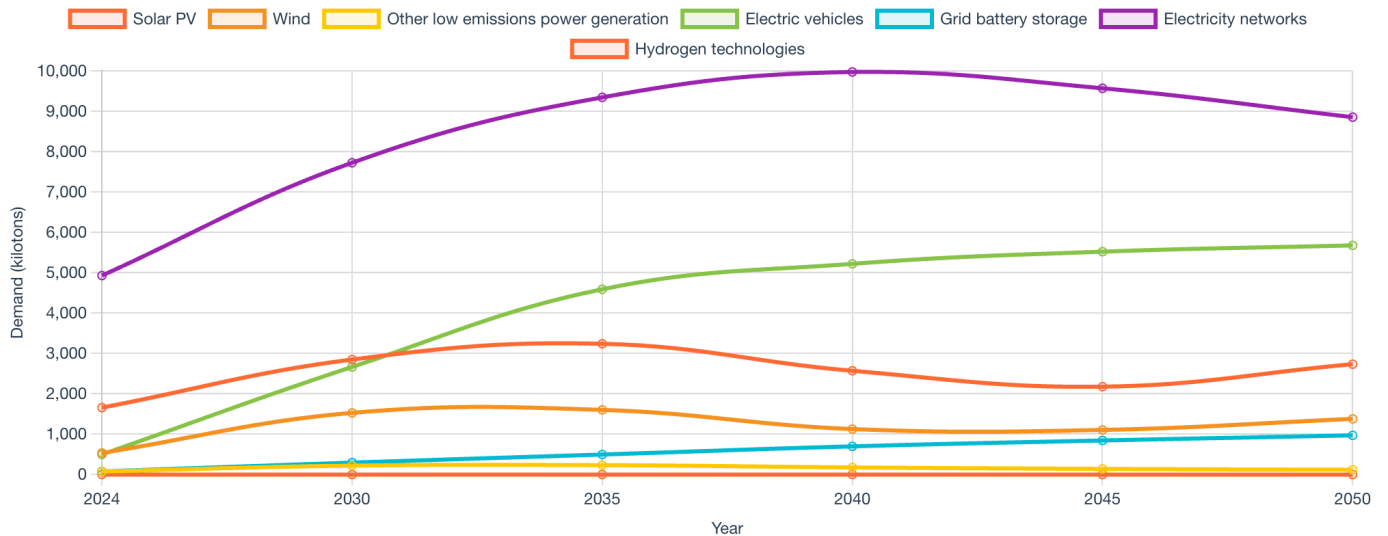
EV lithium (109 → 1547
kt)

 Supply Risk

HIGH

 Sustainability Chatbot

Copper Demand by Technology (Net Zero 2050)



Critical Insights from Real IEA Data

Dramatic Growth in Clean Technology Mineral Demand:

Now: In 2024, total clean technology mineral demand is about 12.6M kt across all tracked minerals.

Future: By 2050, this could increase to 42.9M kt in the Net Zero 2050 scenario.

This represents a **3.4x increase**, highlighting the massive scale-up required in mineral supply chains.

Copper as the Cornerstone of Energy Transition:

Now: Copper demand for clean technologies is about 7.8M kt in 2024.

Future: This could reach 19.7M kt by 2050 (2.5x growth).

Copper is crucial across all clean technologies, with electricity networks consuming 8,852.873 kt by 2050.

Electric Vehicle Revolution Driving Lithium Demand:

Now: Lithium demand for EVs is around 109 kt in 2024.

Future: This could skyrocket to 1547 kt by 2050 in the Net Zero 2050 scenario.

This **14.2x increase** underscores the critical need for expanded lithium supply chains.

Potential Supply Bottlenecks:

Now: Production is highly concentrated - DRC produces ~66% of cobalt, while China refines ~78% of cobalt and ~95% of graphite.

Future: By 2040, demand for many minerals will exceed current production capacity in most scenarios.

This suggests urgent need for investment in new production and supply chain diversification.

Technology Evolution Impact:

Different scenarios show vastly different outcomes - Net Zero 2050 requires 30M kt additional capacity compared to 2024.

Battery chemistry changes could significantly alter demand patterns, while renewable technology improvements may reduce material intensity per unit.

Navigate between scenarios above to see how policy choices dramatically impact mineral demand trajectories and supply chain requirements.

About This Project

Created for the **Viz Con 2025** contest at **Analyticon**(Amazon-wide Conference on Data Engineering, Science and Analytics). This interactive application explores the connections between human activity and environmental conservation through compelling data narratives using real datasets.

Data Sources

-  **Temperature Data:**
NASA GISTEMP v4 (Goddard Institute)
-  **Renewable Energy:**
IRENA Global Energy Statistics
-  **Critical Minerals:**
IEA Critical Minerals Data Explorer
-  **Sea Ice Data:**
NSIDC Sea Ice Index
-  **CO2 Data:**
NOAA Mauna Loa Observatory

Technology & AI

- Frontend:** React.js, D3.js, Chart.js, Framer Motion
- Backend:** Python Flask, NumPy, Pandas
- AI Integration:** Pattern recognition, correlation analysis, predictive modeling, and natural language insight generation
- AI Chatbot:** Amazon Bedrock (Nova Pro Model), API Gateway, AWS Lambda, S3, IAM
- Deployment:** AWS Amplify with automated CI/CD pipeline

Viz Con 2025 Entry

- Entry for the **2025 Viz Con contest**, part of the **Analyticon** - Amazon-wide Conference on Data Engineering, Science and Analytics
- Challenge:** "Visualizing a Sustainable Tomorrow"
- Approach:** Multi-dataset integration with real NASA GISTEMP, NOAA, IRENA, and IEA data
- Goal:** Inspire climate action through compelling data storytelling
- Features:** Interactive visualizations, correlation analysis, predictive modeling, and accessibility-first design

Built with  by [@ygal](#) and his best friend Amazon Q 
Powered by  | NASA, IRENA, IEA, NSIDC & NOAA data